Econometrics III

Module Handbook 2018

Module Code: EC3062

Leicester School of Business

Module Leader: Cheng Chou

Econometrics III EC3062

2018

Contents

Teaching Staff....3
Module Aims....4
Intended Learning Outcomes....4
Delivery Mechanisms....5
Reading....6
Lectures....7
Week 11:....8
Week 12:....8
Week 13:....8
Week 14:....8
Week 16:....8
Week 17:....8
Week 18:....8
Week 19:....8
Week 20:....8
Computer Tutorial....9
Seminars....9
Seminars as a delivery mechanism....9
Role of the student....9
Role of the tutor....9
Assessment....11

This handbook is intended for the guidance of students taking this module in 2018. Whilst the details contained in this handbook represent teaching staff intentions at the time of writing, it is in the nature of higher education that some module information, such as syllabus, reading lists and assignments, *may be subject to modifications during the teaching of a module*. Teaching staff reserve the right to make such minor changes in the matters covered by this publication and will endeavour to publicise any such changes as widely and timely as possible.

Teaching Staff

Module Leader:
Cheng Chou
Office: TBA
Office Hours: Tuesday 15:00-17:00
Email: cchou@le.ac.uk
Office telephone: 0116 373 6288

Two GTA:
Harvey, Laura A.
lah39@leicester.ac.uk
Ghosh, Arpita
ag467@leicester.ac.uk

Your first point of contact with respect to queries and questions is the Econometrics III BlackBoard page. If you cannot find what you are looking for there, then please feel free to contact Cheng Chou via email cchou@le.ac.uk or during office hours.

Any feedback regarding the course, its content or functioning **should be addressed to Cheng Chou**.

Module Aims

1. Master a number of econometric models that can be used to analyse economic data of various structures.
2. An ability to work with big economic survey data (these datasets typically have millions, sometimes billions, of observations and thousands of variables) using the most popular data analysis software called R.
3. An ability to interpret and evaluate critically numerical results of econometric estimation.

Intended Learning Outcomes

On successful completion of this module you will be able to independently find and make a good use of relevant economic survey data to understand certain interesting economic questions as well as their applications in the business world.

Such an ability is particularly valuable in the current technology-driven economy.

Delivery Mechanisms

The module is delivered through a mixture of lectures, seminars and student-centred learning as follows:

Week	Lecture (2 hours)	Computer Tutorial (1 hour)	Seminar (1 hour)
11 (Oct 1)	1, 2		
12 (Oct 8)	3	1	
13 (Oct 15)	4	2	
14 (Oct 22)	5	3	
15 (Bank Holiday)			
16 (Nov 5)	6	4	
17 (Nov 12)	7	5	
18 (Nov 19)	8	6	1
19 (Nov 26)	9	7	2
20 (Dec 3)	10	8	
21		9	

- Computer tutorial aims to teach you how to implement the econometric methods taught in the previous lecture.
- Seminars will review most econometrics methods discussed in the term.

Self study	Continuous	80 hours

The expectation is that all students will **prepare in advance** of each computer tutorial and seminar, having completed all the required reading and any specified preparatory work.

Electronic Resources

Please use the EC3062 site on BlackBoard to access lecture slides, reading materials, seminar preparation work, assessment documents, important announcements and other student resources.

Lecture slides will be posted on Blackboard 48 hours prior to the lecture; this means you will be able to print them if you so wish.

Reading

The **main textbook**, from which the core material is drawn, will be:

- Wooldridge, Jeffrey M. (2015), *Introductory Econometrics: A Modern Approach*, 6th Revised Edition, CENGAGE Learning Custom Publishing.
- Matloff, Norman (2011), The Art of R Programming, O'Reilly Media, Inc.

Recommendation list:

- Wickham, Hadley and Garrett Grolemund (2016), *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*, O'Reilly Media, Inc. [Read free online version here: http://r4ds.had.co.nz]
- Zeileis, Achim and Christian Kleiber (2008), *Applied Econometrics with R*, Springer.
- Chang, Winston (2012), *R Graphics Cookbook*, O'Reilly Media, Inc.

Long, James Scott (2008), *The Workflow of Data Analysis Using Stata*, Stata Press. [This book is about Stata, but it discussed many good practices about doing data analysis.]

Lectures

The two-hour lecture will be delivered once a week with the exception of the first week (Week 11). There will be two lectures in Week 11.

The following outline is tentative. As the course progresses, the lecturer reserves the right to make necessary changes in the module outline.

Week 15 is absent for bank holiday.

Week	Lectures
11	*Two lectures* this week. Course organization, review of linear regression with cross-section data, and introduction to R.
12	Introduction to survey data via IPUMS.
13	Review of simple panel data models
14	Introduction to National Longitudinal Studies
16	Time series analysis and ARIMA model: Basic concepts
17	Time series analysis and ARIMA model: Estimation and Forecast
18	Regression analysis with time series data
19	Some applications of time series analysis: Model selection issues
20	Dynamic panel data models

Week 11:
This session will introduce the module, review the linear regression in cross-section data, and introduce **R** (https://cran.r-project.org). Most of you may not be familiar with R, but mastering R and preferably Python is necessary for any professional involving data analysis in business world.

Week 12:
Unlike the econometrics modules you had, this module will teach how to run an empirical economic analysis from scratch. That means you will NOT be provided data. It's up to you to find and prepare data. This is usually challenging and involves a lot of computer programming. I will show you how to get started with IPUMS --- the single most popular survey data portal in the world.

Week 13:
Review of panel data models and how to implement them in R.

Week 14:
We have learned a few panel data models. We are going to show how to get panel data from a large-scale economic survey --- National Longitudinal Studies.

Week 16:
You have learned cross-section data in detail and have learned basic panel data models. The third common data structure is time series, such as the gasoline price over the past decade. We talk about the basic concepts that would help our understanding about time series.

Week 17:
We introduce one popular time series model --- ARIMA model and discuss its estimation techniques. Moreover, we are going to see how to predict future using ARIMA model, e.g. how to predict the gasoline price in the next month.

Week 18:
ARIMA model can be used analyse the time series about one variable, such as gasoline price. When we have both gasoline price and oil price, how can we learn their relationship using linear regression? It turns out this is tricky.

Week 19:
You will see that one can propose a variety of econometric models to explain time series. How to pick one? We will walk through the step by an example.

Week 20:
When we add time series into classical panel data models, we will have dynamic panel data models. It turns out that its estimation is nontrivial.

Computer Tutorial

You will have a one-hour computer tutorial every week. By working with R software, the tutorial will help you to gain insights about the econometric methods/concepts taught in the previous weeks.

Prior to attending the lab session, it is important that you complete any preparatory work that can be found on BlackBoard.

Please note: success on this module is determined by your willingness to engage and work hard.

Seminars

In Week 18 and 19, you will have a one-hour seminar. Seminar will summarize the econometric concepts taught in the term. Some assignments will be posted to help your understanding of some key ideas.

Please check your timetable for your seminar time and room.

Seminars as a delivery mechanism

Seminars are large group sessions within which small group work and discussion take place. They are your chance to test ideas, raise issues, challenge assumptions and discuss differences in perception. They are designed as a safe environment in which to engage in directed learning. Students will typically work in groups of 5-6 people. Ideas, questions and conclusions developed in the small groups will tend to be shared (at the end of sessions) with the entire seminar. This means that students benefit from the intensity of small group discussion and the knowledge generated by a larger forum. Seminars start from the basis that all statements are open to question, and any opinion may be put forward as long as they can be substantiated, and do not cause deliberate offence on the grounds of age, disability, race, gender or sexual orientation.

Role of the student

The success of any seminar is dependent on the input of its participants. Key in this respect is the student. To get the most out of the seminars (to have meaningful discussions and learning experiences) you will need to undertake the core reading and have completed any preparatory work prior to attending the seminar.

Role of the tutor

Each session is led by a tutor. Their role is to set the ground rules for each session, describing the aims and broad structure. They also work the room providing assistance and guidance during set exercises/discussions. They will provide information where possible, or direct to sources of information where they do not have the answers themselves. They are there to **guide** discussions, **not dominate them**. They will bring

the session to a conclusion, drawing out key themes and their relation to the learning objectives. They will, where possible, seek to answer questions relating to wider aspects of the course.

Assessment

You will need to submit two items --- (1) a group **empirical research paper**, and (2) a **proposal** for that research. See the deadlines on the next page.

After week 15, you will need to submit a project proposal that must include the following elements:

a) a description of your economic question,
b) the data source,
c) the descriptive statistics of a few key variables, and
d) the econometric methods/models to be applied.

<u>What is "empirical research"?</u>

"Empirical research" in economics involves collecting data and running econometric analysis of the collected data in order to understand certain economic question. One sample paper can be found here:
https://www.aeaweb.org/articles?id=10.1257/aer.103.3.251
By the end of the term, your group will submit a similar paper like the linked file. Of course, your topics, data, and methods can differ from the cited paper above.

<u>Assessment of your submitted proposal:</u>

As long as your proposal cover all four required items, you will get the full mark. Each item accounts for 25 percent. So, don't miss any of them.

<u>Assessment of your submitted paper:</u>

The assessment will be based on the novelty and significance of the topic studied (10%), data manipulation (30%), econometrics methodology (30%), interpretation of results (10%) and overall writing (20%).

For <u>an excellent paper</u> (with a grade higher than 85%), I would be looking for an interesting topic with solid data and appropriate econometrics method. Moreover, the presentation should be concise and accurate, and the interpretation should be able to show your understanding about the econometrics methods used.

<u>Penalty for free-ridders:</u>

Because this is a group project for econometrics, every group member should contribute to data manipulation and econometric analysis. Your grade will be significantly **penalized** for the lack of contribution to your team.

<u>Resit:</u>

If you failed the module (mark less than 40%) then you will have the opportunity to resit. For the resit, you do not need to submit proposal but have to complete the project **alone.**

Deadlines:

Please submit both items online through the portal on BlackBoard before the following date.

Item	Hand-in Date	Weighting	Feedback Date
Proposal	Fri, Nov 2, 2018 15:00 UK Time	20	Tue, Nov 6, 2019
Group empirical research paper	Tue, Jan 8, 2019 at 15:00 UK Time	80	Tue, Jan 15, 2019

General reminder for submissions

- The only permitted way that you can submit your assignment is via Blackboard and the online submission process.
- **DON'T leave it till the last minute to submit your work.** Assessment deadlines often coincide with other programmes in the department and across the wider University, the increased activity affects the response time of Blackboard slowing it down. The portal closes at the specified time and you will not be able to submit your work.
- Any assignments received after this deadline will incur the relevant late penalty as follows: (i) a deduction of 10% of the maximum mark available from the actual mark achieved by the student shall be imposed upon expiry of the deadline; (ii) a further deduction of 5% of the maximum mark available from the actual mark achieved by the student shall then be imposed on each of the next subsequent working days.
- All resubmissions (except successful MC claim submissions) will be subject to a 40% cap.
- Mitigating Circumstances If you have an unavoidable difficulty that prevents you submitting on time, such as a sudden illness, then the School are able to consider mitigating circumstances. Please see the below link with detailed information on mitigating circumstances. Please remember you need to provide evidence to support your reasons for being unable to submit the coursework.
  https://www2.le.ac.uk/offices/sas2/regulations/mitigating-circumstances

Mark	Undergraduate Grade Descriptor
85-100%	**Scholarship**: Excellent application of a rigorous and extensive knowledge of subject matter; perceptive; demonstrates a critical appreciation of subject and extensive and detailed critical analysis of the key issues; displays independence of thought and/ or a novel and relevant approach to the subject; reveals both breadth and depth of understanding, showing insight and appreciation of argument. **Independent learning**: Work draws on a wide range of relevant literature and is not confined to reading lists, textbooks or lecture notes; arguments are well supported by a variety of means. **Writing skills**: Writing skills are excellent; writing is clear and precise; arguments are logical, well-structured and sustained, and demonstrate thorough understanding; conclusions are reasoned and justified by evidence. **Analysis:** Work demonstrates a robust approach to analysis that is evident of a deep understanding of relevant concepts, theories, principles and techniques. For quantitative modules analysis is complete and entirely relevant to the problem.
70-84%	**Scholarship**: Very good application of a rigorous and extensive knowledge of subject matter; demonstrates a critical appreciation of subject; displays detailed thought and consideration of the subject; reveals very good breadth and depth of understanding. **Independent learning**: Work draws on a range of relevant literature and is not confined to reading lists, textbooks or lecture notes. **Writing skills**: Writing skills are well-developed; writing is clear and precise; arguments are logical, well-structured and demonstrate thorough understanding; conclusions are justified by evidence. **Analysis:** Analytical steps carried out carefully and correctly demonstrating that it is based on a sound understanding**.** Analysis is relevant to the problem and is complete and is placed in a clear context.
60-69%	**Scholarship**: Good, broad-based understanding of subject manner; makes effective use of understanding to provide an informative, balanced argument that is focussed on the topic; reveals some attempt at creative, independent thinking; main points well covered, displaying breadth or depth but not necessarily both; broadly complete and relevant argument; **Independent learning**: Sources range beyond textbooks and lecture material and are used effectively to illustrate points and justify arguments. **Writing skills**: Arguments are presented logically and coherently within a clear structure and are justified with appropriate supporting evidence; capably written with good use of English throughout; free from major errors; complex ideas are expressed clearly and fluently using specialist technical terminology where appropriate. **Analysis:** Some minor slips in the steps of the analysis and some minor gaps in understanding of underlying principles**.** Analysis is relevant to the problem and mostly complete. A good interpretation which conveys most of its meaning.
50-59%	**Scholarship**: Some but limited engagement with, and understanding of, relevant material but may lack focus, organisation, breadth, and/or depth; relatively straightforward ideas are expressed clearly and fluently though there may be little or no attempt to synthesise or evaluate more complex ideas; exhibits limited independent creative thought; adequate analysis but some key points only mentioned in passing; arguments satisfactory but some errors and perhaps lacking completeness and relevance in parts. **Independent learning**: Sources may range beyond lecture material and textbooks though effective engagement with and use of the wider literature is limited. **Writing skills**: The question is addressed in a reasonably clear, coherent and structured manner but some sections may be poorly written making the essay difficult to follow, obscuring key points or leading to over-generalisation; competently written with a good use of English throughout (few, if any, errors of spelling, grammar and punctuation). Answers that have upper second-class qualities may fall into this category if they are too short, unfinished or badly organised. **Analysis:** Minor slips and occasional basic errors in analysis. Underlying principles are mostly understood, but clear gaps are apparent. Analysis falls short of completeness and is a little irrelevant in place but a reasonable interpretation which goes some way to convey its meaning
40-49%	**Scholarship**: Minimum acceptable level of understanding; extremely basic and partial understanding of key issues and concepts; some material may be used inappropriately; uninspired and unoriginal; relies on limited knowledge; analysis poor or obscure, superficial or inconsistent in places; arguments incomplete, partly irrelevant or naive. **Independent learning**: Sources restricted to core lecture material with no evidence of wider reading. **Writing skills**: Though errors may be few and generally insignificant, answer may be poorly focussed on the question, lack rigour and/or consist of a series of repetitive, poorly organised points or unsubstantiated assertions that do not relate well to one another or to the question, although some structure discernible; borderline or poor competence in English (some problems of spelling, punctuation and grammar that occasionally obscures comprehension). **Analysis:** Some knowledge of the analysis to be followed, but frequent errors. Some attention paid to underlying principles, but lacking in understanding and frequently irrelevant. Some interpretation is given, but it does not place the analysis in any real context
40-44%	**Scholarship**: Minimum acceptable level of understanding; extremely basic and partial understanding of key issues and concepts; some material may be used inappropriately. **Independent learning**: Sources restricted to core lecture material with no evidence of wider reading.

	Writing skills: Though errors may be few and generally insignificant, answer may be poorly focused on the question, lack rigour and/or consist of a series of repetitious, poorly organised points or unsubstantiated assertions that do not relate well to one another or to the question; borderline competence in English (some problems of spelling, punctuation and grammar that occasionally obscures comprehension).
30-39%	*Minimum requirements have not been met.* **Scholarship**: Inadequate knowledge of relevant material; omission of key ideas/material; significant parts may be irrelevant, superficial or factually incorrect; inappropriate use of some material; mere paraphrasing of course texts or lecture notes; key points barely mentioned; very weak grasp or complete misunderstanding of the issues; inclusion of irrelevant material; does not address the topic or question. **Independent learning**: Restricted to a basic awareness or no awareness of course material and textbooks; very meagre use of supporting material or unsupported assertions; use of irrelevant or unconvincing material. **Writing skills**: Unacceptable use of English (i.e. comprehension obscured by significant and intrusive errors of spelling, punctuation and grammar); poor and unclear, or totally incoherent, structure. Answers that 'run out of time' or miss the point of the question may fall into this (or a lower) class. **Analysis:** Erroneous analysis with mistakes. Very little attention paid to the underlying principles of the analysis. Far from complete with little relevance to the problem. Limited interpretation that reveals little, if anything, about the meaning
20-29%	**Scholarship**: Displays a superficial appreciation of the demands and broad context of the question but is largely irrelevant, fundamentally flawed, or factually incorrect; inappropriate use of material; mere paraphrasing of course texts or lecture notes; key points barely mentioned; complete misunderstanding of the issues; inclusion of irrelevant material. **Independent learning**: Restricted to a limited awareness of basic course material; unsupported assertions; use of irrelevant or unconvincing material. **Writing skills**: Minimal structure, though may only list key themes or ideas with limited comment or explanation. **Analysis:** Analysis has very significant omissions demonstrating little understanding of problem or underlying principles. Analysis may be ill suited to problem. Very little interpretation of meaning of the analysis.
10-19%	**Scholarship**: No recognition of the demands or scope of the question and no serious attempt to answer it. Complete misunderstanding of the issues; inclusion of irrelevant material. May have simply failed to address the question/topic set. **Independent learning**: No evidence that the most basic course material has been understood; unsupported assertions; use of irrelevant or unconvincing material. **Writing skills**: Without structure; comprehension may be completely obscured by poor grammar, spelling, punctuation. **Analysis:** Virtually complete failure to carry out analysis. No evidence of understanding of underlying principles and bears no relevance to the problem. No attempt to interpret or explain the meaning of the analysis.